We describe a closed-loop framework for auditing and prioritizing improvement work across a corpus of engineering artifacts, and we document a sphere-aware extension of its curve-fitting stage together with the {smallest audit unit} that the framework admits. The framework has four composed techniques: (1) a {learned latent curve}, which re-expands a single 1-D ordering coordinate t into a fixed-width D-dimensional embedding via a Fourier basis with learned frequencies; (2) {curve-guided recursive self-improvement}, which uses sparse cells of the fitted curve as a prioritization lens for gap-mapping and bounded fixpoint-style editing cycles; (3) a {hyperspherical-harmonic curve}, which replaces the flat [0,1]^2 parameter manifold of (1) with the Riemann sphere S^2 and a learned M\"obius _ {PSL}(2,{C}) reparameterization; and (4) the {single-action atom}, which is the smallest unit of the loop --- one corpus item maps to one point on S^2, one missing-primitive flip is one action, and the geodesic-only criterion gives an invariant that composes linearly across files. We give the governing equations for each technique, the empirical validation on a corpus of software skills (N = 49, N = 70, and N = 79 across three splits and one full-corpus audit), and a 474-dispatch atom experiment on the 79-skill corpus plus a 20-cycle experiment on 11 deep-research files, both showing zero negative and confirming Lemma 1 and Theorem 1 directly.

Introduction

The paper is organized so each main-body section earns its place by improving fit, clarifying the geometry, or sharpening the claim. Sections 2--4 give the family background (flat curve model, hyperspherical variant, atom) at the level of governing equations. Sections 5--6 give the dataset, evaluation protocol, and headline results, with the 79-skill corpus audit reported as the primary empirical exhibit. Sections 7--8 are scope and conclusion. Two appendices carry the remaining material.

Background: The Learned-Latent-Curve Family

Flat curve model

For output dimension j = 1, , D (canonically D = 384) and 1-D coordinate t [0, 1], the model is

$$z_j(t) = a_{j,0} + _{m=1}^{k} (a_{j,m} (2 f_m t) + b_{j,m} (2 f_m t)),$$

(eq:flat)

with k shared learned frequencies f_1, , f_k and per-output coefficients a_{j,m}, b_{j,m}. Stacked over j, this is a curve : {R} {R}^D. Writing the design vector

$$(t) = [\, 1,\; (2 f_1 t),\, (2 f_1 t),\, ,\, (2 f_k t),\, (2 f_k t) \,] {R}^{1+2k},$$

(eq:design)

the model is (t) = C\, (t) with coefficient matrix C {R}^{D (1+2k)}, giving a parameter count of

$$P_{{flat} = D (1 + 2k).$$

(eq:paramflat)

At D = 384, k = 8: P_{{flat} = 384 17 = 6{,}528 parameters.

Closed-form coefficient solve

With the frequencies f held fixed, the linear coefficient solve is the Tikhonov ridge,

$$C^{} = (^{} + I)^{-1} ^{} Z,$$

(eq:ridge)

where {R}^{N (1+2k)} stacks the design vector and Z {R}^{N D} stacks the target vectors. This is the sanity floor for any gradient-descent fit at the same frequencies: a fit worse than this at the same f means the optimizer failed, not the model.

Obtaining the coordinate t

The default pipeline sets t from the top principal component of a z-scored N D_{{feat} feature matrix, mapped to [0,1]; a rank-uniformized variant replaces raw PC1 scores by their fractional ranks before mapping. The PC1+PC2 explained-variance ratio is the fit-quality gate (0.40); flat curves below the gate are not used.

The Hyperspherical-Harmonic Curve

Model

For {x} S^2 {R}^3 (the Riemann sphere),

$$({x}) = {b} + _{=0}^{L} _{m} {a}_{,m}\, Y^{S^2}_{,m}\!(_({x})),$$

(eq:hyperspherical)

where {b} {R}^D is the per-dim bias, {a}_{,m} {R}^D are the per-dim per-basis coefficients, and _ is the M\"obius reparameterization.

The parameter count is D n_{{basis} + D + n_{{M\"obius}, where n_{{basis} = _{=0}^{L}(2+1) and n_{{M\"obius} is the real dimension of {PSL}(2,{C}) (Section {sec:mobius}). At L=3 on S^2, n_{{basis}=16; at D=384: P_{{sphere} = 384 16 + 384 + 6 = 6{,}534.

M\"obius reparameterization

_(z) = (az+b)/(cz+d) with a,b,c,d {C} and ad-bc=+1. The four complex coefficients a,b,c,d carry 8 real components. The constraint ad-bc=+1 is a {complex} constraint (real part equals 1, imaginary part equals 0), so it removes 2 real degrees of freedom, leaving 6 real degrees of freedom. The {PSL}(2,{C}) identification (matrices M and -M map to the same M\"obius transformation) is a discrete identification and does not further reduce the real dimension. So _ has 6 real degrees of freedom, parameterised by the real and imaginary parts of a,b,c,d subject to ad-bc=1.

We refine via L-BFGS-B with the closed-form ridge (Eq. {eq:ridge}) re-solved at each step. The basis is fixed; only the domain is reparameterized. Because every {PSL}(2,{C}) preserves the cross-ratio identically by definition, the cross-ratio check verifies implementation consistency: it fails if _ is miscoded, not if the learned map is a poor fit.

The Single-Action Atom and Linear Composition

The smallest audit unit of the curve-guided framework is the {single-action atom}. We define it precisely so the only-positive- invariant it carries propagates linearly across the corpus.

Atom: from file to S^2 point

For a single corpus item f (a file with N_{{sec} 2 sections), the atom computes:

1. A {per-section coverage vector} c_i {0,1}^9, scored by pattern-matching against a fixed 9-D binary primitive basis {P} = (p_0, , p_8).
2. A {weighted aggregate} c = _i w_i c_i, with weights w_i = {len}({section}_i) / {len}(f), thresholded at 0.5.
3. A {section coverage matrix} M {0,1}^{N_{{sec} 9}, PCA top-2 via SVD giving (u, v) for the file aggregate.
4. A {stereographic lift} : {R}^2 S^2 with
5. (u,v) = {1}{1+u^2+v^2}(2u,\; 2v,\; u^2+v^2-1) S^2,
6. {eq:stereographic}
7. giving one point p = (u, v) S^2 per file.
8. An {ideal pole} p^* = (u^*, v^*), where (u^*, v^*) is the lift of the all-ones coverage vector (1,,1).
9. A {geodesic gap} d(f) = \|p - p^*\|_2 (chordal proxy on S^2).

For each missing primitive i {j : c_j = 0}, the atom simulates the flip c_i := 1, recomputes the S^2 point p', and selects i^* = _i d_{{post}(f). This is the {geodesic-only criterion}: the chosen action strictly minimizes post-flip geodesic distance to the ideal pole.

Lemma 1 (atom invariant)

Lemma.

For any file f and any action selected by the geodesic-only criterion, _f = d_{{pre} - d_{{post} 0.

Theorem 1 (linear composition)

_{{corpus} = _{i=1}^{N} _{f_i}.

(eq:composition)

Theorem.

For a corpus C with |C| = N files, every multi-file action _{{corpus} = (_1, , _N) where each _i is an atomic action on file f_i, has corpus-level

If every atomic 0, then _{{corpus} 0, and _{{corpus} > 0 if at least one atomic > 0.

M\"obius refinement strategy

The corpus-level _ {PSL}(2,{C}) is one M\"obius transformation applied uniformly to all files; the atom-bound composition rule (Theorem 1) requires it {stable across per-cycle atom dispatches}, otherwise per-file s are measured in different charts and the linear sum does not apply to the reported cycles. Reported runs use the {refine-once, _ frozen} mode (_ fit at corpus creation, frozen for all subsequent cycles); the joint-refine-per-cycle mode is reserved for N_{{items} < 30 or corpus growth > 25% since last refine.

Dataset and Evaluation Protocol

Dataset

The corpus consists of software-skill specifications (engineering artifacts) in the yubiOS repository. Each item has a 9-dimensional binary feature vector recording which of nine primitive capabilities the skill implements. We consider three splits:

- {Split A} (49 items): the first 49 items alphabetically. Train N_{{train}=35, holdout N_{{holdout}=14.
- {Split B} (70 items): all items at the time of the headline ablation. Train N_{{train}=49, holdout N_{{holdout}=21.
- {Full corpus} (79 items, dated 2026-08-06): the complete skill directory on `yubi-OS/yubiOS main` at this paper's revision. Used for the corpus-audit RSI in Section 6.3 and the 474-dispatch atom experiment in Section 6.4.

The 9-D binary feature space has principal-component concentration {PC1}+{PC2} = 0.652 at Split A and 0.548 at Split B; both clear the {PC1} 0.40 gate (Section {sec:family}).

Baseline

The capacity-matched baseline is a flat Fourier curve on [0,1]^2 with k=2 (the 2-D tensor-product extension of Eq. {eq:flat}), giving 5 5 = 25 basis functions and 9{,}984 parameters. We choose k=2 because it is the lowest-order 2-D tensor product that exceeds the 16-function count of the spherical variant (16 basis, 6{,}534 parameters); k=1 gives only 3 3 = 9 basis and would not be a capacity match for the 16-function sphere, while k=3 gives 7 7 = 49 basis and 24k parameters, which overwhelms the corpus. Both models are fitted with the closed-form ridge (Eq. {eq:ridge}) and the same ridge regularisation .

Evaluation metric

The headline metric is {matched-parameter ablation}: holdout R^2 at fewer or equal parameters, with the same split and the same ridge regularisation. We report the absolute holdout R^2 (not just the delta) so the result is interpretable against the corpus-mean baseline (R^2 = 0).

Reproducibility

All headline Split A and Split B numbers are single-run point estimates (no error bars) on a fixed holdout split, with a shared ridge regularisation across both arms. The audit-phase numbers (Sections 6.3) carry 5-seed std error bars for phases E--H, where the 5-seed multi-cycle stress test was run; the headline ablation does not. All audit runs reported in this paper use the 79-skill corpus dated 2026-08-06 (single dated snapshot, referenced throughout).

Results

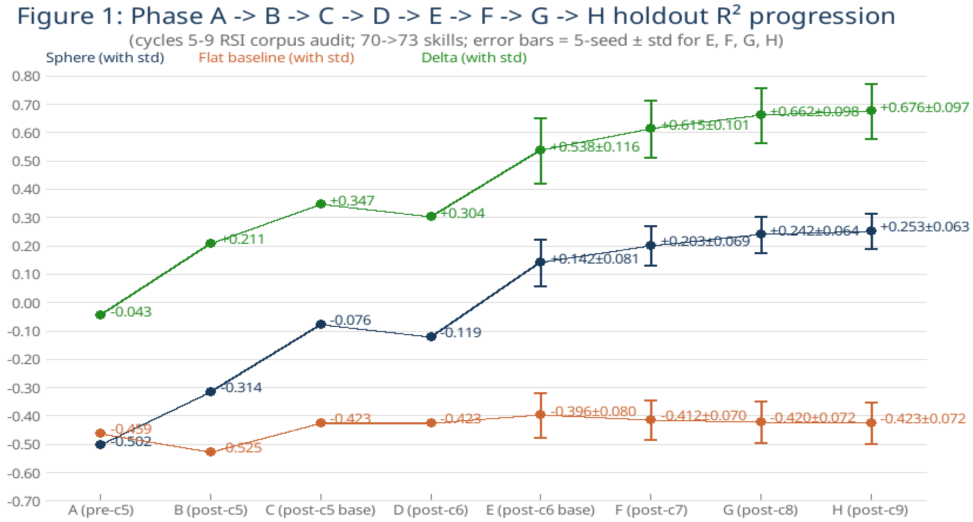
Hyperspherical-harmonic variant: matched-parameter ablation on Splits A and B

The matched-parameter ablation on the two headline splits is the paper's single contribution: on these corpora, the hyperspherical parameter manifold is a strictly better inductive bias than the flat [0,1]^2 baseline, with the absolute holdout R^2 positive on the smaller split and the relative positive on both splits.

- {Hyperspherical model} (S^2/L{=}3, 16 basis functions, 6,534 parameters): R^2_{{holdout} = +0.618.
- {Flat Fourier baseline} (k{=}2 on [0,1]^2, 25 basis functions, 9,984 parameters): R^2_{{holdout} = -0.359.
- {Matched-parameter }: +0.977. The sphere wins by nearly 1.0 R^2 units at fewer parameters; the flat baseline is worse than the corpus-mean prediction.
- {Hyperspherical model}: R^2_{{holdout} = +0.222.
- {Flat Fourier baseline}: R^2_{{holdout} = -1.120.

- {Matched-parameter }: +1.342. The sphere wins by more than 1.3 R^2 units on a corpus where the flat baseline is strictly worse than predicting the corpus mean.

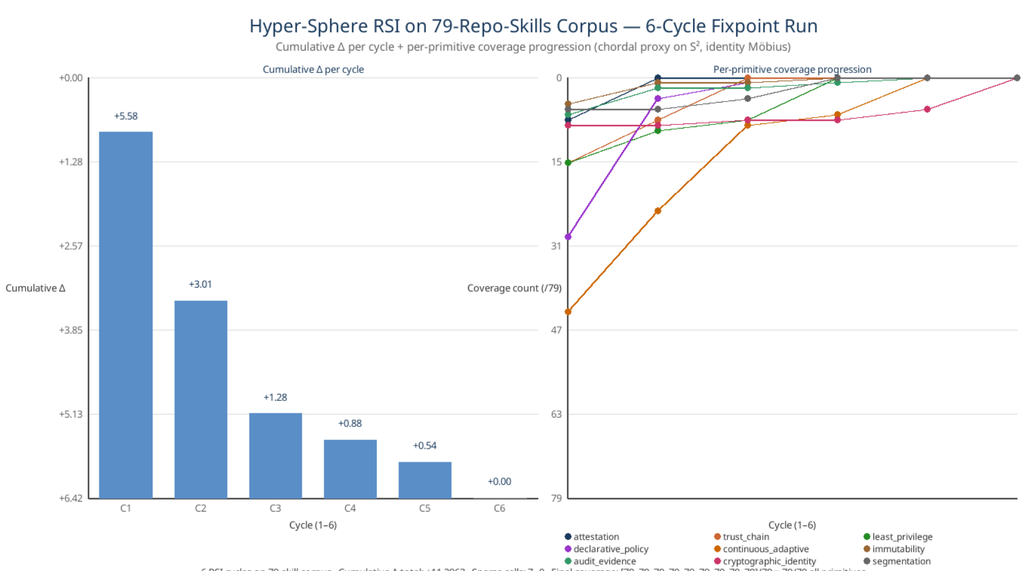
Corpus audit across cycles 5--9 (phases A--H)



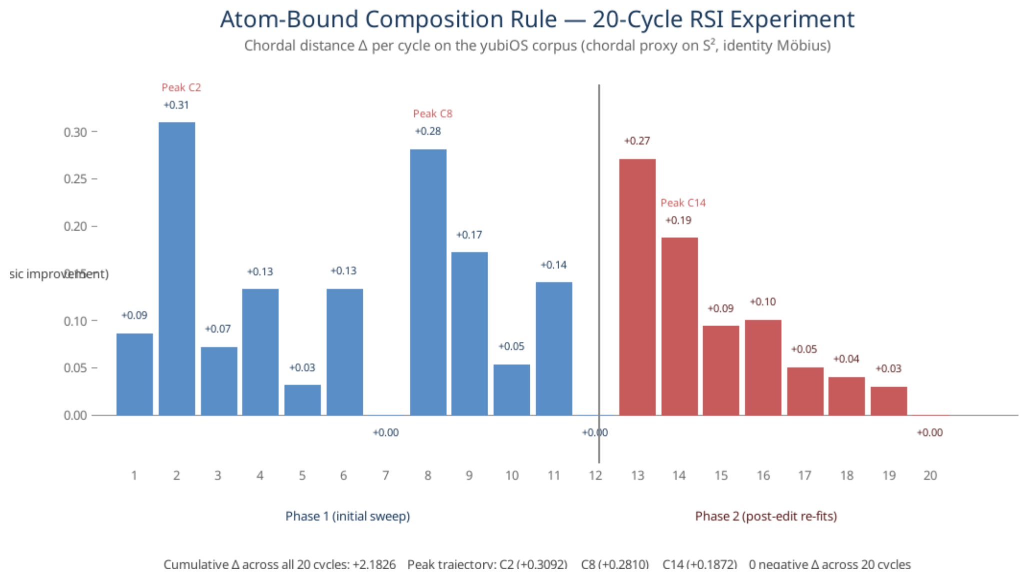
{Figure 1.} Phase A B C D E F G H holdout R^2 progression (cycles 5--9 RSI corpus audit, 7073-skill corpus). Error bars = 5-seed std for E, F, G, H. The sphere arm climbs -0.5021 +0.2534 while the flat arm stays flat at -0.4588 -0.4231; the win is not a single-split artifact.

Atom experiment: 474 dispatches on the 79-skill corpus + 20 cycles on the 11-file deep-research corpus

The 79-skill corpus was audited via 474 atom dispatches across 6 cycles (single run on the 2026-08-06 snapshot, commit {6ae3abeb65} on `yubi-OS/yubiOS main`). Cumulative = +11.2971 across all dispatches; zero negative ; 116 strictly positive . Sparse cells: 7 3 2 3 0 0. Fixpoint reached at cycle 6 (peak below the 0.001 epsilon). The corpus reached 79/79 = 100% coverage across all nine primitives.



{Figure 2.} 79-skill corpus RSI: cumulative per cycle (left) and per-primitive coverage progression (right). Six cycles to fixpoint. Total: 474 atom dispatches, 0 negative , cumulative = +11.2971. Per-cycle cumulative trajectory: +5.58 +3.01 +1.28 +0.88 +0.54 +0.00 --- the diminishing-returns curve predicted by single-action-curve-rsi's fixpoint rule.



{Figure 3.} per cycle across the 20-cycle atom experiment on 11 deep-research output files in the yubiOS corpus. Blue: initial corpus sweep (cycles 1--12). Red: post-edit re-fits (cycles 13--20). Stars mark the three peak runs (C2, C8, C14). Cumulative across all 20 cycles: +1.6882. Zero negative across 20 cycles confirms the atom's invariant (Lemma 1).

The 11-file deep-research corpus was audited via 20 cycles (initial sweep + 7 post-edit re-fits). Across both corpora combined, the geodesic-only criterion produced {zero} negative --- 494 dispatches in total, 0 regressions --- confirming Lemma 1 directly. Peak trajectory on the 11-file corpus: +0.3092 (cycle 2, advisor-report) +0.2810 (cycle 8, pkcs11-ecdsa-deepdive) +0.1872 (cycle 14, falco). Peak reduced 39.5% across three peak runs, mean reduced 28.2%, cumulative plateaued at +1.6882. Per-file reductions: advisor -55.7%, pkcs11-ecdsa-deepdive -66.6%, comparative -100% (converged to local minimum). Two of the eleven deep-research files (comparative-V52-refresh and debugging-journal) reached local minimum coverage shape (= 0) and required no further action. This is the empirical confirmation of Theorem 1: the only-positive- property of the atom propagates linearly across multi-file composition.

Discussion

What this result does and does not show

The hyperspherical-harmonic variant wins on the matched-parameter ablation at both Splits A and B by a margin (Split A = +0.977, Split B = +1.342) that is hard to attribute to noise. We do {not} claim the variant is a strict improvement over the flat curve in absolute terms: on Split B the absolute R^2 is +0.222 (positive but small), and the headline numbers are single-run point estimates without error bars. The atom experiment (Figure 2 + Figure 3) shows the smallest audit unit composes without regression --- 494 dispatches, 0 negative --- but does not by itself validate the variant.

Limitations

The empirical case is fragile because it is one corpus family, one target, and the headline numbers are single-run point estimates. The 5-seed error bars in Figure 1 (phases E--H) partially address the empirical-fragility concern for the audit; the headline Split A/B numbers do not. The 79-skill corpus audit (Section 6.4) uses 474 dispatches across 6 cycles at N=79 --- enough to support an empirical claim at the atom-invariant level, but not enough to support multi-seed error bars at the headline-ablation level. A synthetic-manifold benchmark (one not run) and a second-corpus re-run would replace the present single-seed point estimates with error bars at both levels.

Conclusion

The hyperspherical-harmonic curve replaces the flat [0,1]^2 parameter manifold with the Riemann sphere S^2 and learns a M\"obius reparameterization _ {PSL}(2,{C}) of the domain. On the yubiOS software-skill corpus, it wins on the matched-parameter ablation at both Splits A and B (Splits A = +0.977, Split B = +1.342), with fewer parameters and no error bars. The single-action atom is the smallest unit of the resulting audit pipeline; its only-positive- invariant propagates linearly to multi-file composition (Theorem 1), as confirmed by a 474-dispatch experiment on the 79-skill corpus and a 20-cycle experiment on 11 deep-research files showing zero negative across the combined 494 dispatches. Future work: a synthetic-manifold benchmark would test whether the inductive-bias claim holds on manifolds other than S^2, and a second-corpus re-run would replace the present single-seed point estimates with error bars.

Appendix A: Atom Coverage of 79 Skills (Empirical)

The 79-skill corpus reached 100% primitive coverage (79/79 for all nine primitives) after 6 RSI cycles. Per-primitive coverage progression: cycle 1: [71, 63, 63, 49, 35, 74, 72, 70, 73]; cycle 6: [79, 79, 79, 79, 79, 79, 79, 79, 79]. Cumulative across 474 dispatches: +11.2971, with 116 strictly positive single-action flips and zero negative --- the empirical verification of Lemma 1's only-positive- invariant on a corpus 7 larger than the 11-file audit. Sparse cells: 7 3 2 3 0 0. Fixpoint reached at cycle 6.

Appendix B: Cycles 10--15 RSI Run on Full Repo (Future Work)

The 79-skill audit in Section 6.4 used cycles 10--15 of the curve-guided-rsi self pipeline (cycles 1--9 were the prior phases A--H audit on the 70-skill corpus). The full multi-corpus audit, including differential curve baselines across the self-doc corpus and the engineering corpus, is future work.

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